Height Growth Equations for CONUS

Greg Johnson, David Marshall, Aaron Weiskittel

2026-03-03

Data

We took the FIA height growth data, filtered out FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 3223563 growth observations.

The observations are distributed among species codes as follows (pre:

Species	Common Name	n
131	loblolly pine	319728
316	red maple	217592
202	Douglas-fir	151818
318	sugar maple	122671
746	quaking aspen	115776
12	balsam fir	91766
611	sweetgum	87177
802	white oak	86252
122	ponderosa pine	71333
241	northern white-cedar	70707
108	lodgepole pine	66643
621	yellow-poplar	62048
129	eastern white pine	60524
833	northern red oak	56635
125	red pine	56026
261	eastern hemlock	49675
111	slash pine	45109
531	American beech	44781
762	black cherry	43400
95	black spruce	42925
263	western hemlock	42011

Species	Common Name	n
110	shortleaf pine	39808
375	paper birch	38339
832	chestnut oak	37641
19	subalpine fir	36013
97	red spruce	35433
827	water oak	34510
68	eastern redcedar	34077
835	post oak	34059
837	black oak	33031
543	black ash	32019
693	blackgum	30729
371	yellow birch	29173
93	Engelmann spruce	28789
541	white ash	28403
951	American basswood	27351
15	white fir	26229
972	American elm	25997
544	green ash	25690
17	grand fir	23933
71	tamarack (native)	23603
743	bigtooth aspen	22791
409	mockernut hickory	21784
403	pignut hickory	21733
132	Virginia pine	20497
94	white spruce	20038
711	sourwood	18688
372	sweet birch	18207
812	southern red oak	18058
971	winged elm	18005
121	longleaf pine	17563
105	jack pine	17322
806	scarlet oak	17080
823	bur oak	16517
11	Pacific sliver fir	16379
407	shagbark hickory	15852
242	western redcedar	15511
264	mountain hemlock	14025
701	eastern hophornbeam	13151
602	black walnut	11337
462	hackberry	9509
694	swamp tupelo	9413

Species	Common Name	n
402	bitternut hickory	9374
313	boxelder maple	9314
631	tanoak	9103
820	laurel oak	8996
741	balsam poplar	8959
931	sassafras	8933
73	western larch	8802
591	American holly	8785
408	black hickory	8637
81	incense-cedar	8105
809	northern pin oak	7730
351	red alder	7702
653	sweetbay magnolia	7542
317	silver maple	7530
391	American hornbeam, musclewood	7295
491	flowering dogwood	6991
975	slippery elm	6433
98	Sitka spruce	5996
901	black locust	5968
831	willow oak	5230
116	Jeffrey pine	5106

Height Growth Prediction

We fit the equation:

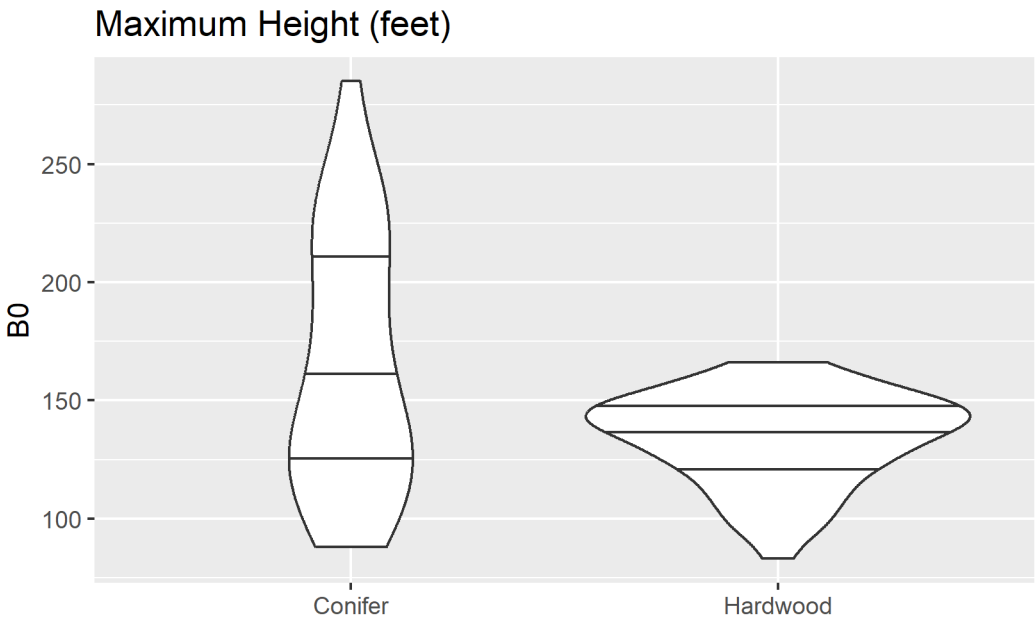
$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

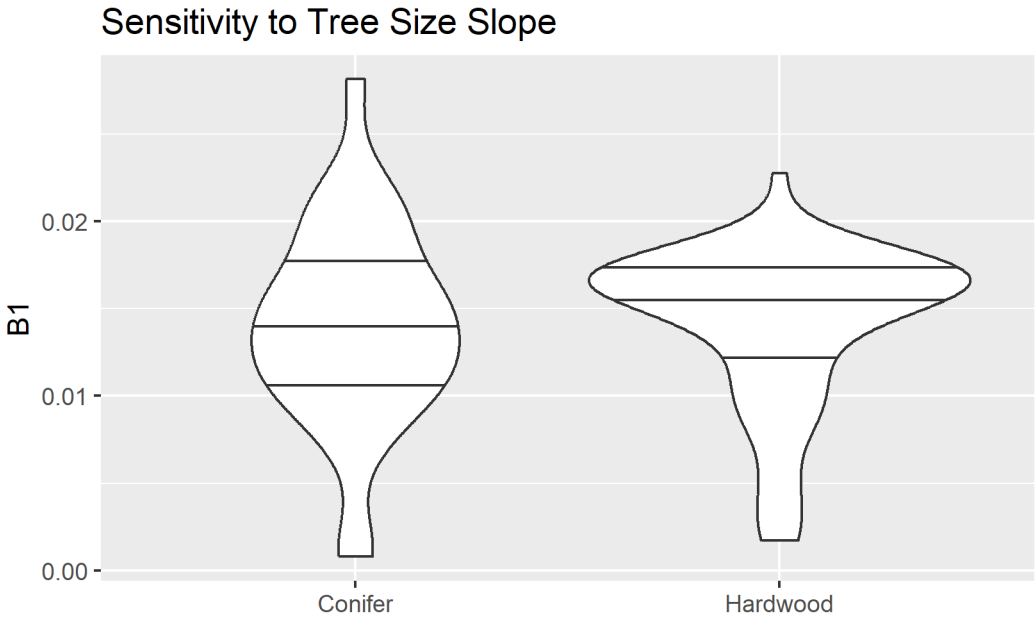
to the all 3223563 growth observations, estimating β_0 through β_6. Where:

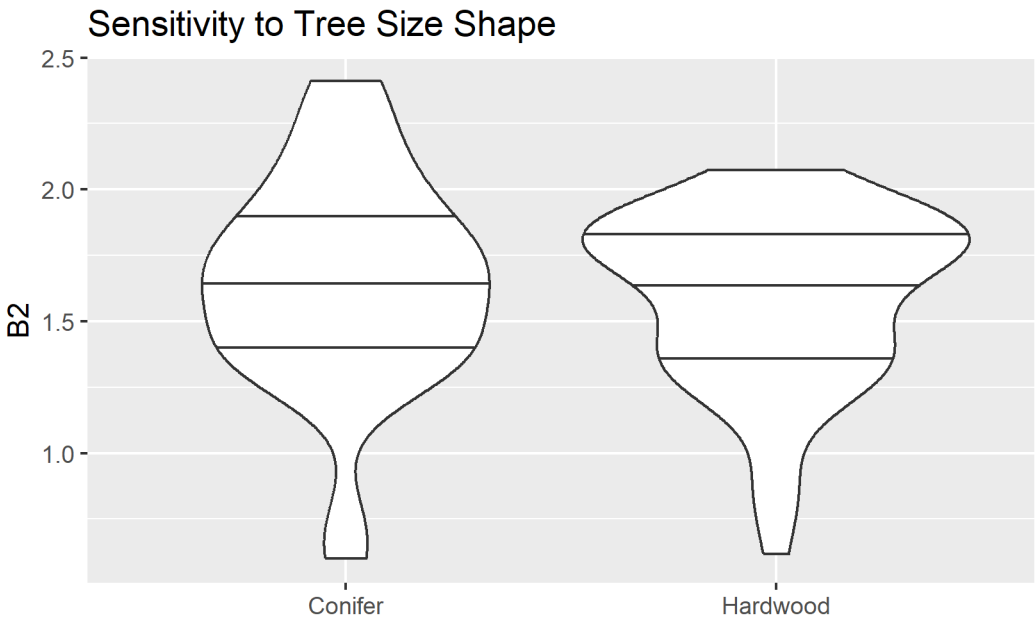
- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

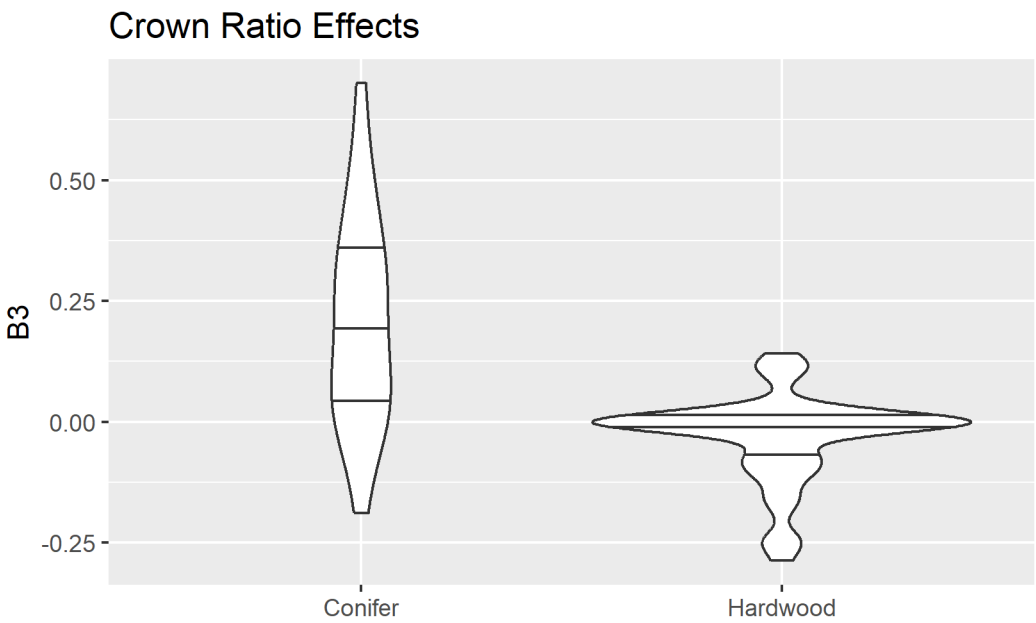
ht_{max} was estimated by finding the maximum height observed in the data set for each species.

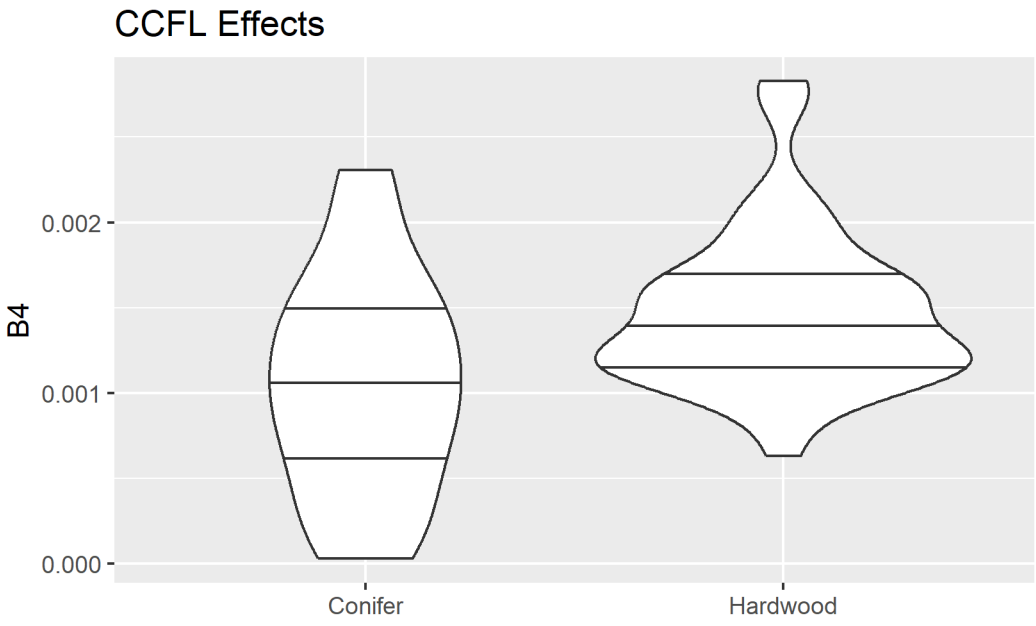
Preliminary CONUS-wide Height Growth Equation

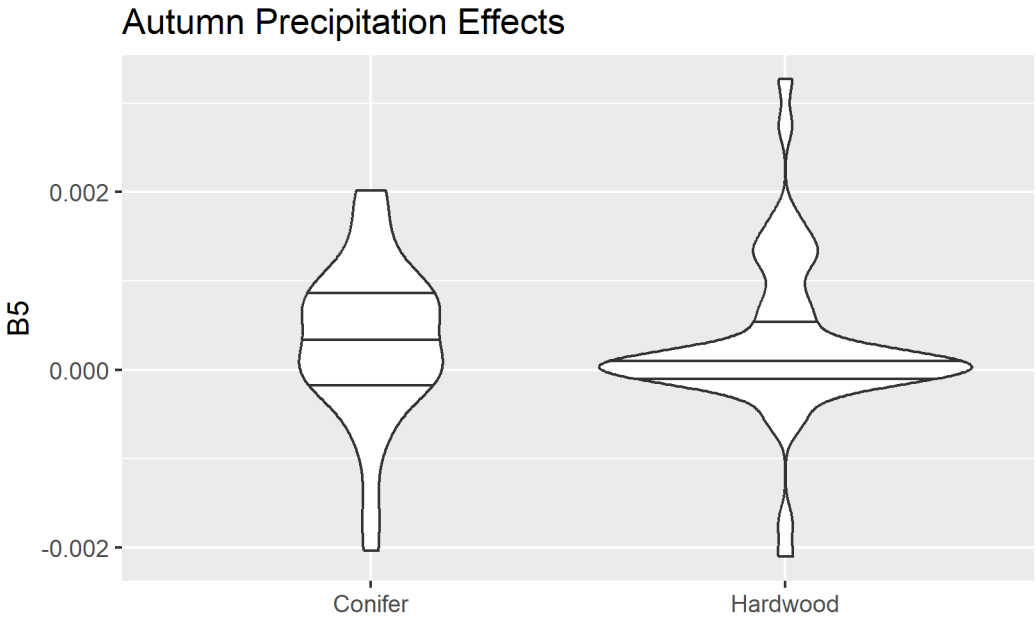


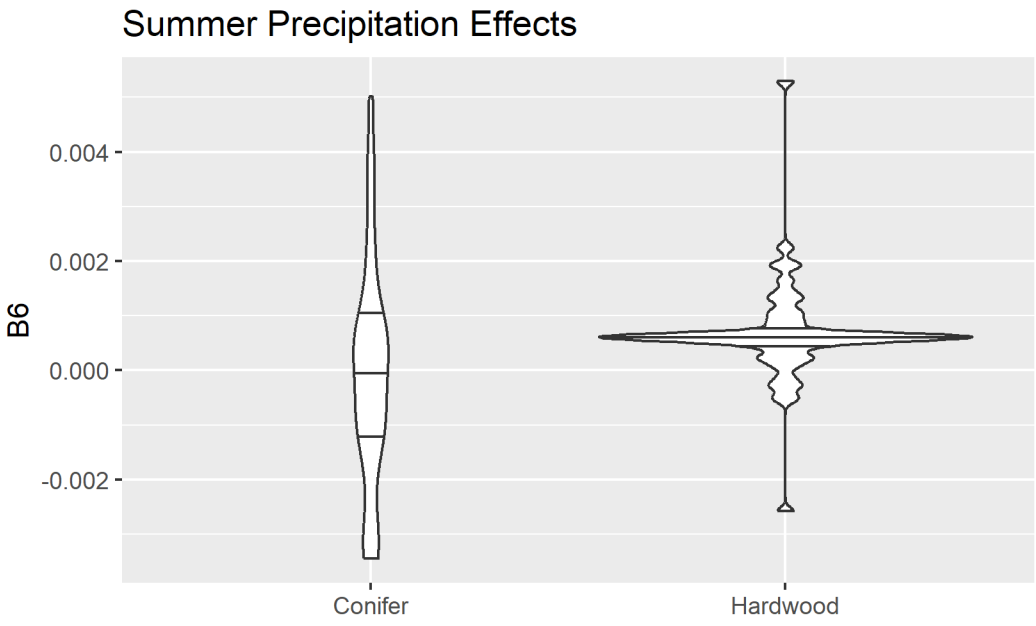


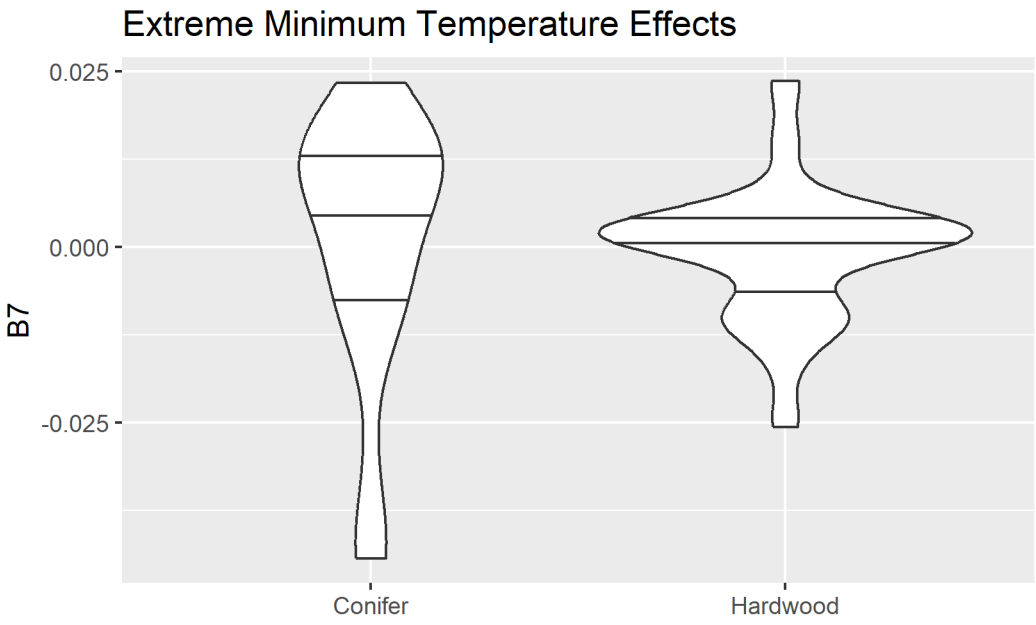

Sensitivity to Tree Size Shape
B2
2.5
2.0
1.5
1.0
Conifer
Hardwood

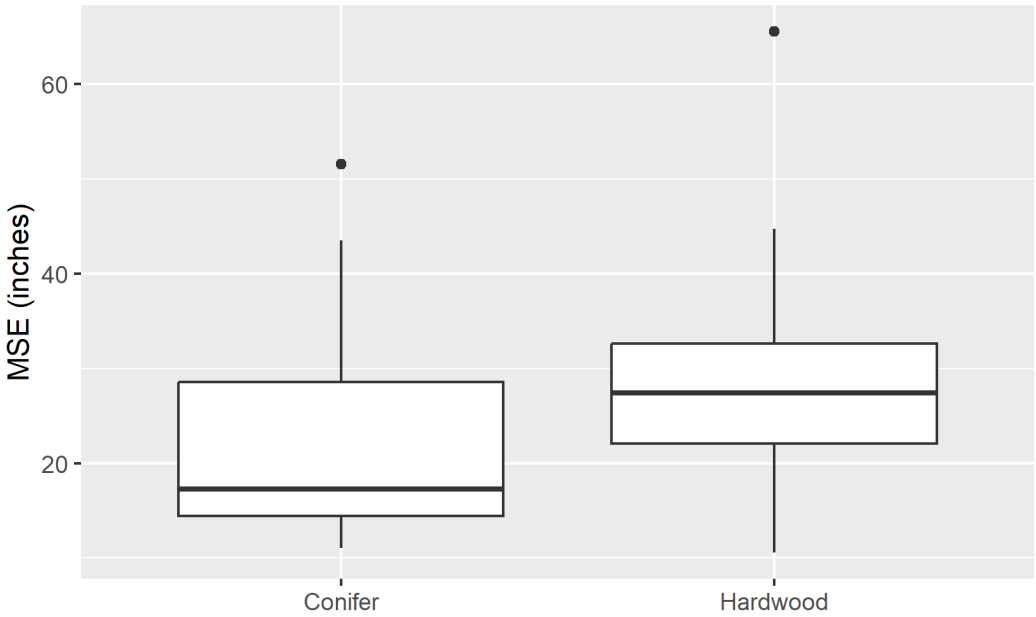

Crown Ratio Effects
B3
0.50
0.25
0.00
-0.25
Conifer
Hardwood


CCFL Effects
B4
0.002
0.001
0.000
Conifer
Hardwood


Autumn Precipitation Effects
B5
0.002
0.000
-0.002
Conifer
Hardwood


Summer Precipitation Effects
B6
0.004
0.002
0.000
-0.002
Conifer
Hardwood


Extreme Minimum Temperature Effects
B7
0.025
0.000
-0.025
Conifer
Hardwood



FIA Species Clusters by Height Growth Parameter Estimates

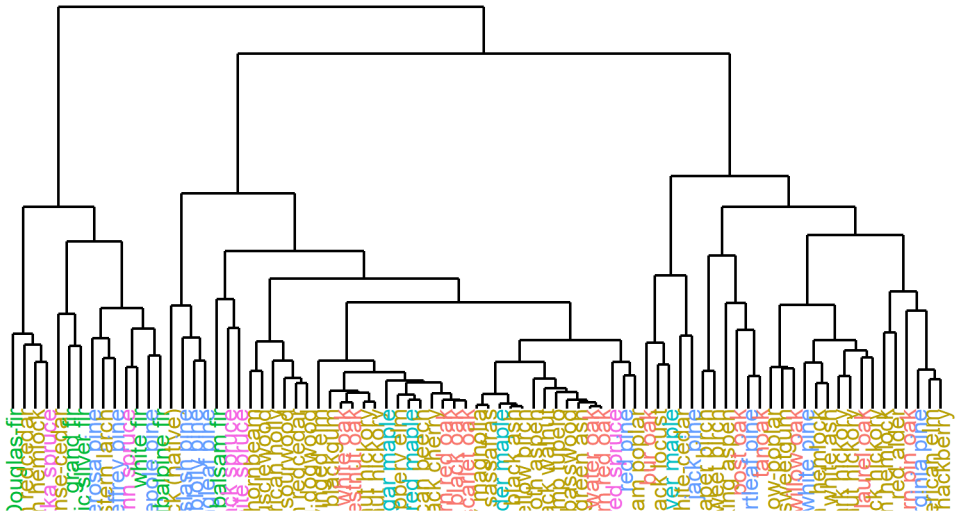


Table 2: Species with Negative Crown Ratio Parameter

FIA Species Code	Common Name
241	northern white-cedar
375	paper birch
261	eastern hemlock
541	white ash
820	laurel oak
701	eastern hophornbeam
391	American hornbeam, musclewood
105	jack pine
372	sweet birch
823	bur oak
591	American holly
407	shagbark hickory
264	mountain hemlock
741	balsam poplar
831	willow oak
901	black locust
408	black hickory
402	bitternut hickory
317	silver maple
631	tanoak
351	red alder
809	northern pin oak

Table 3: Species with Negative Autumn Precipitation Parameter

FIA Species Code	Common Name
746	quaking aspen
318	sugar maple
12	balsam fir
110	shortleaf pine
694	swamp tupelo
375	paper birch
129	eastern white pine
832	chestnut oak
95	black spruce
693	blackgum
68	eastern redcedar
491	flowering dogwood

FIA Species Code	Common Name
125	red pine
97	red spruce
743	bigtooth aspen
711	sourwood
971	winged elm
71	tamarack (native)
409	mockernut hickory
105	jack pine
94	white spruce
372	sweet birch
591	American holly
741	balsam poplar
631	tanoak
81	incense-cedar

Table 4: Species with Negative Summer Precipitation Parameter

FIA Species Code	Common Name
202	Douglas-fir
12	balsam fir
241	northern white-cedar
122	ponderosa pine
108	lodgepole pine
263	western hemlock
132	Virginia pine
97	red spruce
93	Engelmann spruce
701	eastern hophornbeam
71	tamarack (native)
15	white fir
823	bur oak
264	mountain hemlock
462	hackberry
242	western redcedar
901	black locust
402	bitternut hickory
351	red alder
98	Sitka spruce
116	Jeffrey pine

Table 5: Species with Negative Extreme Minimum Temperature Parameter

FIA Species Code	Common Name
131	loblolly pine
611	sweetgum
746	quaking aspen
12	balsam fir
111	slash pine
621	yellow-poplar
241	northern white-cedar
694	swamp tupelo
129	eastern white pine
261	eastern hemlock
972	American elm
541	white ash
125	red pine
132	Virginia pine
97	red spruce
121	longleaf pine
820	laurel oak
743	bigtooth aspen
71	tamarack (native)
105	jack pine
372	sweet birch
823	bur oak
407	shagbark hickory
741	balsam poplar
831	willow oak
462	hackberry
242	western redcedar
901	black locust
408	black hickory
402	bitternut hickory
317	silver maple